

Nikolaos (Nikos) Triantafyllou

PHYSICIST, PHD CANDIDATE IN COMPUTATIONAL ASTROPHYSICS AND COSMOLOGY

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Research Interests

Large-scale structure, 21cm cosmology, machine learning, bayesian statistics

Education

Scuola Normale Superiore

Pisa, Italy

PHD IN COMPUTATIONAL ASTROPHYSICS AND COSMOLOGY

Nov. 2023 - Nov. 2027 (Est.)

- Advisor: Prof. A. Mesinger

University of Crete

Heraklion, Greece

MSc GRADUATE DEGREE IN ADVANCED PHYSICS (SPECIALIZATION: ASTROPHYSICS)

Sept. 2022 - June 2023

- Grade: 8.88/10 ("Excellent")
- Advisor: Prof. V. Pavlidou

University of Crete

Heraklion, Greece

BSc UNDERGRADUATE DEGREE IN PHYSICS

Sept. 2018 - Sept. 2022

- Grade: 7.95/10 ("Very Good"; top 3% among all Department of Physics graduates).
- Advisor: Prof. V. Pavlidou

Research Experience

Scuola Normale Superiore

Pisa, Italy

SUPERVISOR: PROF. ANDREI MESINGER

2023 - Present

- Description: Research on Large-scale structure and Cosmology for the EoR. Using **bayesian** techniques (Posterior Density Estimation, Neural Ratio Estimation, Diffusion Models) for **field-level, multi-tracer** inference of the **primordial density field** and galaxy properties at $z > 5$.

University of Crete and IA FORTH

Heraklion, Greece

SUPERVISOR: PROF. VASILIKI PAVLIDOU

2021 - 2023

- Description: Research on Large-scale structure and Cosmology. Used deep (NNs, **CNNs**, GraphNNs) and shallow learning to probe the **turnaround radius** on the plane of the sky based on mass and line-of-sight velocity simulated data at $z < 1$ to constrain cosmological parameters.

First author Publications

A full publication list can be found here: <https://nikos-triantafyllou.github.io/research.html>

IN PREPARATION

TILING I: Field-level Bayesian Reconstruction of Cosmological Initial Conditions during the Epoch of Reionization

N. Triantafyllou, A. Mesinger, S. Murray, S. Gagnon-Hartman

TILING II: Inferring the build-up of Reionization and unseen Galaxies

N. Triantafyllou, A. Mesinger

21cmVAULT: Multi-tracer Inference of Galaxy Properties for the First Billion Years of the Universe and an Extensive Database

N. Triantafyllou, A. Mesinger

PUBLISHED

Searching for a Signature of Turnaround in Galaxy Clusters with Convolutional Neural Networks

(<https://arxiv.org/abs/2412.13304>)

N. Triantafyllou, G. Korkidis, V. Pavlidou, P. Bonfini

International Conferences

Presentations can be found here: https://github.com/nikos-triantafyllou/conference_talks/
For all conferences, see: <https://nikos-triantafyllou.github.io/conferences.html>

A New Era in Astrophysics: preparing for early science with the SKAO

SPEAKER

Görlitz, Germany

June 2025

Simulation-Based Inference for Galaxy Evolution

SPEAKER

Bristol, UK

May 2025

Teaching Experience

Spring 2025	Structure Formation in the Early Universe , Creation of a hands-on tutorial for cosmological initial conditions	Pisa, Italy
2020 - 2022	Three labs: Advanced Physics, Optics and Electromagnetism , Teaching Assistant (6 months for each)	Heraklion, Greece

Awards/Fellowships

April 2026	Compute Time Allocation for AI & Machine Learning project: "Effect of field-level-conditioning on cosmological and astrophysical inferences" , LEONARDO HPC/CINECA	95,0000 standard hours
September 2025	Compute Time Allocation and a physical Nvidia L4 GPU winner of poster session. , E4 Computer Engineering	Nvidia L4 GPU
January 2025	Compute Time Allocation for AI & Machine Learning project: "Reconstruction the Initial Conditions of the Universe using Diffusion Neural Networks" , LEONARDO HPC/CINECA	80,0000 standard hours
2018-2019	"Chrysanthos and Anastasia Karidis" Bequest Scholarship ,	

Skills

Programming & Software

Working with: Python, $\text{\LaTeX}$, HPC systems, database creation, parallelization

Familiar with: C, C++, MATLAB, website building

Languages

English (fluent, B2-ECCE, working proficiency), Italian (conversational, B2), Greek (native)

Academic

Creativity, problem-solving, data science, teaching, report & proposal writing, organizing group meetings/journal clubs, presentation & communication skills, work ethic, logo design

Other Interests _____

Music (guitar playing, singing, lyric writing), philosophy, art, movies, road trips, swimming, basketball, wine/beer tastings, bars/pubs

Other Projects _____

AstroVisio (INAF, Design Group Italia)

Project that aims to develop an innovative virtual reality tool to help researchers visualize astrophysical data

Contribution: Explaining the challenges that astrophysicists face in visualizing data, during the very first and intermediate stages of the project. Provided astrophysical data from simulations. Created a video demo for the possible use cases of Astrovisio for a relevant presentation of the company.